

Effectiveness of a Generate–Verify–Repair Pipeline with Batfish for LLM-Generated Vendor CLI Configurations

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Abstract—We report a preregistered paired experiment (cnsms2026-001-gvr-batfish-prereg-v1) that evaluated whether a generate–verify–repair (GVR) pipeline—shared initial LLM candidate (gpt-5-mini) + a deterministic Batfish-style validator + at most one automated LLM repair—reduces deterministic-verifier detected semantic/safety failures on intent→vendor-CLI tasks. Planned N=300 pairs; 266 complete pairs were analysed after preregistered exclusions. Baseline pass-rate = 260/266 (97.744%); guarded (final GVR) pass-rate = 265/266 (99.624%). Paired difference = 0.01879699248; exact McNemar two-sided $p = 0.0625$. Paired bootstrap (B=10,000, seed=1729) 95% percentile CI = [0.0037593984962406013, 0.03759398496240601]. Repair was invoked for 6 tasks; median repair iterations per task = 0. Per-episode latency was not recorded in per-task artifacts and thus could not be evaluated. Under shared_initial_candidate semantics the observed absolute improvement is small and did not meet the preregistered $\alpha=0.05$ decision rule. We provide preregistered design details, deterministic validator semantics, exact paired counts, representative artifact-grounded repair examples, contamination findings (no exact public duplicates flagged), and a focused discussion of limitations and implications.

I. INTRODUCTION

Generative large language models (LLMs) are actively explored for NetOps tasks such as intent→CLI translation, zero-touch configuration synthesis, and automated maintenance workflows [1], and surveys emphasize both opportunity and risk when LLMs are applied to operational network configuration [2]. Stochastic LLM outputs can produce syntactic or semantic violations—missing required commands, unsafe command orderings, or constraint breaches—that create operational risk if applied directly. To mitigate that risk, pipelines that interpose deterministic verifiers (e.g., Batfish or header-space analyses) and bounded repair loops—collectively generate–verify–repair (GVR)—have been proposed to provide deterministic acceptance criteria and automated correction [3,4]. Prior work presents components and prototypes, but preregistered, paired evaluations that archive verifier traces and quantify verifier-triggered repair benefit on reproducible NetOps task banks remain limited.

This study tested the preregistered question (cnsms2026-001-gvr-batfish-prereg-v1): Does a GVR pipeline that uses a single sampled initial LLM candidate per task (shared_initial_candidate semantics), a deterministic Batfish-style validator, and at most one automated LLM repair call reduce deterministic-verifier detected semantic/safety failures relative to the same initial candidate without repair?

Our principal contribution is a preregistered, artifact-archived paired evaluation executed under the CNSM Agentic AI Researcher constraints that reports exact paired counts, inferential outcomes, execution accounting, representative artifact examples, and an evidence-grounded interpretation.

II. RELATED WORK

Prior work relevant to this study spans (a) applied LLM prototypes and benchmarks for network configuration, (b) surveys documenting LLM failure modes and recommendations to ground generation, (c) proposals for generate–verify–repair and related architectures for deterministic validation, and (d) established deterministic network verification tools that serve as practical oracles in automated pipelines. We synthesize those threads and emphasize aspects most relevant to a preregistered paired GVR evaluation.

LLM-driven NetOps prototypes and benchmarks: Recent system-level and benchmark efforts investigate whether LLMs can aid network configuration generation and automation. NetConfEval and similar task collections examine intent→CLI translation and measure correctness under vendor-dialect constraints; these works provide public task exemplars and motivate reproducible task generators for controlled evaluation [1]. Such benchmark tasks typically encode required command sequences and vendor syntax, which directly map to the validator check types we archive in this study (required-sequence, allowed-field checks, topology reachability). Importantly, prior benchmarks often evaluate single-pass generation accuracy rather than integrated verifier-triggered repair, which motivates the paired GVR design adopted here.

Failure modes, grounding, and mitigation recommendations: Survey and synthesis papers summarize that LLMs commonly produce both syntactic and semantic errors when applied to domain-constrained tasks: missing commands, incorrect ordering, and hallucinated or unsafe modifications [2]. Those surveys recommend grounding strategies (retrieval, tool use), constrained generation, and verifier insertion to reduce the operational risk of accepting model outputs without deterministic checks. Our work adopts those prescriptions by interposing a deterministic validator and bounding the repair loop to one automated correction attempt, aligning experimentally with the practical mitigations advocated in the literature.

Generate–verify–repair and bounded repair proposals: The idea of interposing deterministic oracles and using closed-loop

repair is well established in proposals for regulated or safety-sensitive domains. Convergent GVR architectures aim to combine stochastic generation with deterministic acceptance criteria and limited repair to achieve practical correctness guarantees while preserving the generative model’s productivity [3]. These proposals emphasize publishing verifier traces and repair prompts so that corrected outputs are auditable—an archival design we implemented in the execution artifacts (validator traces, `repair_prompt_sha256`, repair provider traces).

Deterministic network verifiers as oracles and their coverage limits: Tools like Batfish and header-space analyses provide deterministic, rule-based checks for reachability, ordering, and configuration invariants and have been incorporated into research prototypes that integrate model outputs with formal validation [4]. Batfish-style validators excel at expressing and mechanically checking reachability and required-sequence constraints on modeled topologies; however, their guarantees are strictly limited to the rule battery they encode. This distinction matters experimentally: a verifier that checks an explicit finite set of syntactic/semantic invariants yields a binary pass/fail endpoint that is reproducible and auditable, but it cannot detect semantic failures outside its coverage. Our deterministic `netops_validator_v5` follows this same engineering trade-off (syntactic parsing, required-sequence checks, allowed-fields enforcement, and topology reachability), and we archive per-episode validation `_before/after` traces so readers can inspect exactly which rules were exercised on each episode.

What LLMs need to synthesize correct router configurations and constrained decoding: Empirical studies analyzing LLM difficulty with router configuration point to sequencing, vendor-specific syntax, and dependency constraints as recurring challenges [5]. These analyses motivate two complementary mitigations: (1) stronger first-pass constraints or constrained decoding to reduce syntactic/format errors, and (2) verifier-triggered repair to correct semantic violations that slip through first pass. Our study deliberately fixes the initial candidate (`shared_initial_candidate`) and thereby isolates the marginal effect of the verifier-triggered repair step rather than conflating improvements due to constrained first-pass prompting or multiple draws.

Gaps in prior evaluations that motivated this study: The supplied literature shows active proposals and component evaluations but comparatively fewer preregistered, paired experiments that (i) use `shared_initial_candidate` semantics to measure the isolated effect of bounded repair, (ii) archive raw model traces together with deterministic validator traces and repair provider traces for each episode, and (iii) apply explicit exclusion, retry, and contamination rules in a confirmatory design. Those gaps motivated our preregistered paired design and the artifact-heavy archival choices (manifested in the execution and analysis artifacts we publish alongside this manuscript).

Methodologically relevant syntheses for experimental design: From the surveyed and prototyped prior work we derived three concrete design choices: (1) use a deterministic validator

battery (Batfish-style) to obtain an auditable binary endpoint; (2) bound the repair loop (≤ 1 automated repair call per failed candidate) to maintain a clear paired estimand under `shared_initial_candidate` semantics; and (3) preregister explicit missingness, retry, and contamination treatments to avoid post hoc analytic choices. Each of these choices has precedent or justification in the cited literature: verifiers as deterministic oracles [4], GVR architectures for regulated settings [3], and benchmark/task construction recommendations that emphasize reproducibility and contamination controls [1,2].

In sum, our related-work synthesis locates this study at the intersection of applied intent→CLI benchmarks, survey-driven mitigation recommendations, GVR methodological proposals, and deterministic verification tooling. The specific empirical contribution is a preregistered, archived paired GVR evaluation under `shared_initial_candidate` semantics that fills a documented gap in artifact-grounded confirmatory assessments of verifier-triggered repair effectiveness.

III. METHODOLOGY

Preregistration and estimand: The experiment followed the preregistered plan `cnsms2026-001-gvr-batfish-prereg-v1` (manifest SHA-256: `46a223492f760950b3d06f475129ce21e52e296c4c2a1e09df3ccd50bec9f891`). The primary estimand is the `paired_success_rate_difference_guarded_minus_baseline` under `shared_initial_candidate` semantics. Under these semantics the identical single sampled initial model candidate per task is used to produce both outcomes: (1) baseline — score the single initial candidate with the deterministic validator; (2) guarded — if that same candidate fails, the pipeline may invoke at most one automated LLM repair call and then rescore with the same deterministic validator. This design isolates the verifier-triggered repair effect for a fixed initial draw and matches the preregistered `execution_contract` (`generation_semantics = "shared_initial_candidate"`).

Task bank, sampling, and inclusion rules: The task bank was deterministically generated to produce $N=300$ intent→CLI pairs, stratified into two vendor-dialect clusters (150 Cisco-like, 150 Juniper-like) via a seeded synthetic generator supplemented with public NetConfEval-style items as specified in the preregistration. Generator ids and deterministic seeds are archived in the task manifest. Inclusion and exclusion rules followed the preregistered contract: public exact-duplicate tasks detected by the contamination check were to be excluded from the primary confirmatory analysis (none were excluded for exact duplication in this run); provider/API transient failures triggered up to two automatic retries, and pairs where both arms failed due to infrastructure/model API errors were excluded from the primary paired analysis per the `'complete_pair_primary'` missingness rule. The execution manifest records `planned_episode_count = 600` ($300 \text{ pairs} \times 2 \text{ arms}$) and `completed_episode_count` and exclusions are reconciled in the deterministic reconciliation artifacts.

Model orchestration and runtime configuration: The declared model used for generation and any repair calls was recorded as `gpt-5-mini` (execution/model_configuration.json declared_model_version = "gpt-5-mini"). Archived configuration fields include `max_output_tokens = 1024`, `maximum_attempts_per_call = 1`, `provider = openai_responses`, and `temperature = null` (unset). Provider accounting recorded `hosted_model_call_event_count = 306`, `completed_hosted_model_call_event_count = 272`, and `failed_hosted_model_call_event_count = 34`; stage accounting recorded `shared_initial_generation = 300` and `repair = 6`. Because temperature was unset and no centralized deterministic sampling seed for provider calls is archived, nondeterministic sampling of provider responses cannot be excluded for the recorded calls; this runtime nondeterminism is declared and discussed in Limitations and the Disclosure Statement.

Deterministic validator design and scoring rules: The binary oracle was `deterministic_netops_validator_v5`. For each candidate it executed a fixed rule battery that is archived per episode: (1) CLI syntactic parsing and normalization; (2) required-sequence checks comparing the task's `required_sequence` to the parsed commands; (3) constrained-field touch checks against `allowed_touched_fields` and `allowed_fields` in the task payload; and (4) reachability/constraint validation over the deterministic synthetic topology model (a lightweight Batfish-style evaluation). Each validator trace archives `validation_before` and `validation_after` objects containing `parsed_command_count`, `required_command_count`, `observed_touched_field_count`, violation codes, `normalized_configuration`, and a boolean valid flag. The confirmatory scoring rule for the primary binary endpoint was `'full_intent_constraint_validation'` as recorded in the scoring artifacts; these artifacts drive the paired contingency counts used in the primary test.

Repair loop mechanics and bounding: Per the preregistration the automated repair stage was strictly bounded to at most one LLM repair call per episode when the initial candidate failed the validator. The repair prompt construction is archived: it includes the task constraints, the validator failure codes and normalized validator diagnostics, and an explicit instruction to produce a corrected configuration satisfying the same task constraints. Repair provider traces and `repair_prompt` identifiers are archived for each repair invocation. Stage accounting indicates 6 repair calls executed in the run; all repair artifacts (repair prompts, repaired outputs, and subsequent validator traces) are included in the evidence bundle and illustrated in representative scoring JSONs.

Analysis plan and inferential procedures: The prespecified primary hypothesis test was the exact McNemar two-sided test on paired binary outcomes at $\alpha=0.05$ (estimator `id` `paired_success_rate_difference_guarded_minus_baseline`). Interval estimation used a paired bootstrap percentile CI ($B=10,000$, `bootstrap_seed = 1729`) for the paired difference to quantify effect magnitude and uncertainty; these bootstrap parameters and resample counts are archived in the

analysis log. Missingness handling for the primary analysis followed the preregistered `'complete_pair_primary'` rule: pairs with both-arm provider/infrastructure failures were excluded from the confirmatory McNemar test; a preregistered sensitivity analysis treats excluded episodes conservatively as failures. Secondary analyses (preregistered) included `median_repair_iterations_per_task` and median end-to-end latency; the latter could not be computed because per-episode timing was not archived. Multiplicity control for formal secondary tests was specified with a Bonferroni adjustment; exploratory analyses were explicitly labeled and not used to support the primary claim.

Execution logging, archival, and auditability: Every episode archived raw model outputs, normalized configurations, `validator_before/after` traces, scorer decision codes, and provider call accounting. Deterministic reconciliation artifacts verify episode accounting (`completed_episode_count`, `failed_episode_count`, `complete_pair_count`) and provider call tallies. Key execution statistics recorded in the reconciliation artifacts are: `planned_episode_count = 600`, `completed_episode_count = 532`, `failed_episode_count = 68`, `complete_pair_count = 266`, `model_calls_used = 306`, `repair_calls = 6`, `cache_hit_count = 0`, `cache_miss_count = 272`. These archived artifacts permit independent verification of the paired counts (`n11`, `n10`, `n01`, `n00`) and replay of repaired episodes for failure-mode inspection.

Design constraints and estimand implications: We emphasize that `shared_initial_candidate` semantics (one sampled initial candidate per task used to evaluate both baseline and guarded outcomes) was selected and executed exactly as preregistered. This choice isolates the incremental effect of a validator-triggered, bounded repair for the observed initial sample; it does not measure the effect of constrained prompting, ensemble first-pass generation, or multiple independent initial draws. Those alternative interventions would require independent-generation arms or additional preregistered conditions that were not executed under this run and therefore are outside the estimand and scope of the reported confirmatory analysis.

IV. RESULTS

Execution accounting and sample construction. The preregistered run planned 600 model-condition episodes ($300 \text{ tasks} \times 2 \text{ conditions}$). The orchestrated pipeline completed 532 episodes and recorded 68 failed episodes (`execution_manifest.json`). After applying preregistered exclusion rules that remove pairs affected by both-arm provider/infrastructure failures, the confirmatory analysis used 266 complete paired units (`analysis/missingness_summary.csv`). Key provider-call accounting from the deterministic reconciliation artifact: `hosted_model_call_event_count = 306` total model call events, of which 272 completed successfully and 34 failed; cache was unused (`cache_hit_count = 0`, `cache_miss_count = 272`). Stage accounting shows that a shared initial generation was produced for every task (`shared_initial_generation = 300`) and

the automated repair stage executed in 6 episodes (repair = 6). All execution-level manifests and per-episode traces are archived to permit replay and audit.

Primary paired outcomes and exact counts. The confirmatory paired contingency (guarded vs baseline) computed on the 266 complete pairs is shown in analysis/paired_contingency_table.csv and summarized here: $n_{11} = 260$ (both baseline and guarded pass), $n_{10} = 5$ (guarded pass, baseline fail), $n_{01} = 0$ (baseline pass, guarded fail), $n_{00} = 1$ (both fail). These margins yield observed success rates baseline = $260/266 = 0.9774436090$ and guarded = $265/266 = 0.9962406015$ with an absolute paired difference of 0.01879699248.

Inferential results: exact McNemar and paired bootstrap CI. Per the preregistered primary test, the exact McNemar two-sided p-value for the observed discordant counts ($n_{10}=5$, $n_{01}=0$) is $p = 0.0625$, which does not meet the preregistered $\alpha = 0.05$ decision rule for rejection. To quantify effect magnitude uncertainty we computed the preregistered paired bootstrap percentile 95% CI with $B = 10,000$ resamples and seed = 1729 (analysis/analysis_log.jsonl); the resulting CI is [0.0037593984962406013, 0.03759398496240601], narrowly excluding zero. Reporting both results preserves the preregistered decision rule while also conveying interval evidence: the discrete nature of McNemar’s exact two-sided test with few discordants can produce conservative p-values even when bootstrap resampling indicates a small positive paired difference with limited sampling variability. Both inferential artifacts (exact p and bootstrap CI with documented seed and resamples) are archived.

Repair invocation behavior and representative repaired episodes. The repair stage executed for 6 of the 300 tasks (stage_counts: repair = 6). Median repair iterations per task across the analysed sample is 0 (majority of tasks required no repair because the shared initial candidate already passed validation). Representative repaired episode: task-000008. The shared initial candidate for task-000008 (execution/responses/task-000008-shared-initial.txt) contained a truncated final token and failed validator checks (validation_before showed parsed_command_count < required_command_count). The automated repair call produced a corrected candidate (execution/responses/task-000008-guarded.txt) whose validator trace (execution/scoring/task-000008-guarded.json) records validation_after.valid = true with no violation codes. Per-episode scoring JSONs record whether a repair was applied, the repair_prompt reference, and the final normalized_configuration accepted by the deterministic validator; all such per-episode repair traces are archived for examination.

Contamination and sensitivity to missingness. The preregistered contamination detector flagged no exact public duplicates (analysis/contamination_summary.csv empty), so no tasks were excluded from the primary analysis on contamination grounds. The preregistered missingness policy excluded 34 both-arm failed episodes from the primary test; a sensitivity analysis that conservatively treats those 34 excluded pairs

as failures (i.e., complete $N = 300$ with both arms set to failure where missing) yields baseline = $260/300 = 0.866667$ and guarded = $265/300 = 0.883333$, but retains the same discordant structure ($n_{10}=5$, $n_{01}=0$) leading to the same exact McNemar $p = 0.0625$. Thus, under the preregistered conservative missingness treatment the formal decision does not change; archived missingness_summary.csv and deterministic_reconciliation.json provide the complete accounting.

Supplementary quantitative artifacts and reproducibility. Analysis artifacts included in the evidence bundle and referenced above provide the reproducible numeric foundations for these results: analysis/condition_summary.csv (baseline: 260 successes, 6 failures; guarded: 265 successes, 1 failure), analysis/paired_contingency_table.csv (n_{11} , n_{10} , n_{01} , n_{00} table), analysis/missingness_summary.csv (counts of excluded and failed episodes), and analysis/analysis_log.jsonl (bootstrap settings: $B=10,000$, seed=1729). deterministic_reconciliation.json documents provider call totals and stage counts that explain model_calls_used = 306 and repair_calls = 6. All files and their hash manifests are archived to enable exact reanalysis or targeted follow-up audits.

V. DISCUSSION

Statistical interpretation and the role of small discordant counts: The paired analysis observed only five discordant pairs favoring the guarded pipeline ($n_{10}=5$) and zero discordant pairs in the opposite direction ($n_{01}=0$). Exact McNemar two-sided testing evaluates the symmetry of discordant counts and, for small integer discordant counts, yields discrete p-values; with $(b,c)=(5,0)$ the exact two-sided $p = 0.0625$ as archived. The preregistered paired bootstrap percentile CI ($B=10,000$, seed=1729) for the paired difference produces a 95% interval that narrowly excludes zero [0.00376, 0.03759]. Both results are reported because they illuminate different aspects of the same data: the exact McNemar p emphasizes a strict null-hypothesis test under paired discrete outcomes, while the bootstrap CI summarizes empirical uncertainty about the magnitude of the paired difference under resampling of the observed complete pairs (bootstrap settings and seed are archived in analysis_log.jsonl). The observed apparent tension ($p>0.05$, CI excluding zero) arises from discreteness and small discordant counts rather than from contradictory raw facts; readers should interpret the pair together: a small positive effect was observed, but it did not meet the preregistered exact-test decision threshold.

Effect size, power, and practical detectability in context: The preregistered power calculation targeted detecting a 0.10 absolute improvement at 80% power with $N=300$ pairs. The observed effect (≈ 0.0188) is far smaller than that target; with complete_pairs = 266 (reduced from planned 300 due to 34 both-arm provider failures) the realized power to detect such a small effect is negligible. In practical terms this experiment demonstrates that when baseline first-pass success is already high ($\approx 97.7\%$), bounded automated repair that can be invoked at most once per failed candidate will rarely be exercised (6

repairs invoked across 300 planned tasks) and thus can only produce small absolute improvements unless the baseline error rate is larger or the repair stage is more permissive (e.g., allow multiple repair iterations or stronger repair strategies).

Artifact-grounded diagnostics of pipeline failure modes: The archived per-episode validator traces permit direct inspection of why candidates failed and how repairs corrected them. Representative diagnostics include: - Truncation/malformed output: task-000008's shared initial candidate shows a truncated terminal line ("interface uplink2\n") that caused `parsed_command_count` mismatches; the automated repair produced the missing explicit admin down line and then passed the validator. Repair artifacts record both the validator violation codes and the repair prompt used, enabling replay of the exact repair stimulus. - Required-sequence violations: several failure traces indicate missing or misordered commands relative to the task's `required_sequence`; these are precisely the semantic checks the deterministic validator enforces and which repair prompts referenced when invoked. - Constrained-field touch checks: validator traces include `observed_touched_field_count` and `allowed_touched_fields` enforcement; failures of this type reflect field-scope violations rather than pure syntax and are detectable deterministically by the validator. These concrete, archived diagnostics show that the validator's rule battery produced meaningful, actionable failure signals that the repair prompt could address in some cases, but also that many initial candidates already satisfied the rule battery and thus were unaffected by the repair stage.

Operational and design implications grounded in execution artifacts: The execution accounting (`deterministic_reconciliation.json` and provider call logs) shows: `hosted_model_call_event_count` = 306, `completed_hosted_model_call_event_count` = 272, `failed_hosted_model_call_event_count` = 34, `shared_initial_generation` = 300, and `repair` = 6. This instrumentation supports several concrete operational inferences: 1) Low marginal utility of bounded repairs on high-accuracy corpora: with few initial failures, a guarded repair stage invoked at most once yields a small absolute improvement; this suggests teams evaluating GVR in practice should calibrate repair budget (number of allowed repair iterations, repair prompt sophistication) to baseline error rates. 2) Auditability and triage: the archived validator traces and scoring JSONs provide a compact, machine-readable audit trail suitable for operator triage and for post-deployment incident analysis; these artifacts concretely demonstrate how deterministic validation can localize a failure cause (syntax vs sequence vs field scope). 3) Latency and throughput unknowns: per-episode timing was not archived and therefore the preregistered latency estimand could not be evaluated; absent timing data, operational latency tradeoffs between invoking a repair and manual review remain unmeasured in this run.

Threats to validity revisited with artifact evidence: Internal validity: the study respected `shared_initial_candidate` semantics preregistered intent and used identical initial candidates

for both arms; as a result, the measured estimand isolates the effect of the validator-triggered repair only. Construct validity: `deterministic_netops_validator_v5` enforces a limited, archived rule battery (syntactic parsing, `required_sequence` checks, `constrained_field` checks, reachability on the synthetic topology). Semantic failures outside that battery (for example, subtle policy violations not encoded in the validator) would go undetected and are thus absent from the measured endpoint. External validity: the task corpus combined public NetConfEval-style examples and a seeded synthetic generator limited to two vendor dialect clusters; the manuscript therefore does not claim generality to other vendors, larger topologies, or production heterogeneity. Conclusion validity: provider failures reduced the analyzed N (266 complete pairs) and the small number of discordant pairs limits inference for small effects.

Concrete recommendations for future experiments (artifact-grounded): The archived manifests suggest the following preregistered changes would increase diagnostic value and power in follow-on studies: (a) design arms that manipulate first-pass generation (e.g., constrained prompting, multiple independent draws) rather than relying solely on `shared_initial_candidate` semantics if the goal is to measure effects of prompting or sampling; (b) allow a bounded sequence of repair iterations (≥ 1) and instrument repair-step efficacy; (c) record per-episode latency (request→generation→validation→repair→final validation) to enable formal latency/operational tradeoff analysis; (d) expand and publish the validator rule battery and add adversarial synthetic tasks specifically crafted to probe known verifier false negatives—per-episode validator traces in this run show that such audits are feasible; and (e) if the target operational environment has high baseline accuracy, power calculations should target smaller effect sizes or alternative estimands (e.g., time-to-triage reduction, human-in-the-loop workload) that better reflect marginal benefits of repair.

Reproducibility and archival diagnostics: All raw responses, provider call traces, validator traces, scoring JSONs, analysis artifacts (paired contingency table, condition summary, missingness summary, contamination summary), `deterministic_reconciliation.json`, `model_configuration.json`, and `analysis_log.jsonl` (bootstrap parameters: $B=10,000$, $seed=1729$) are archived and hashed in the evidence bundle. These artifacts permit exact reanalysis of the paired contingency, replay of repair prompts with the same validator failure codes, and targeted inspection of individual failure modes. The provider accounting and stage counts enable independent auditors to confirm that the repair stage executed only six times and that 34 provider calls failed and were excluded per preregistered rules.

Synthesis: The experiment provides an artifact-rich, preregistered measurement of the limited marginal benefit of a tightly bounded GVR pipeline under `shared_initial_candidate` semantics when baseline first-pass accuracy is high. The archived traces make clear which failure modes the validator detected, which were amenable to one automated repair call,

and where further engineering (expanded validator coverage, richer repair strategies, or different experimental arms) would be required to observe larger absolute benefits.

VI. LIMITATIONS

- Shared_initial_candidate semantics: both arms used the same single sampled initial candidate per preregistration; the estimand measures validator-triggered repair benefit for that fixed initial sample and does not evaluate independent first-pass generations or constrained first-pass prompting strategies.
- Missingness and sample reduction: planned N=300 pairs; after infrastructure/provider exclusions 266 pairs were complete and 34 both-arm episodes were excluded per preregistered rules, reducing effective sample size and precision.
- Small discordant counts: primary inference used exact McNemar with few discordants ($n_{10}=5$, $n_{01}=0$); conclusions are sensitive to inferential choice and limited power.
- Validator coverage: deterministic_netops_validator_v5 enforces a finite set of syntactic/semantic/reachability rules; semantic violations outside its coverage are possible and unmeasured.
- Runtime nondeterminism and sampling: archived execution/model_configuration.json records temperature = null and no deterministic seed; nondeterministic sampling cannot be excluded for recorded model calls.
- H2 latency not evaluated: per-episode end-to-end latency was not present in archived per-task artifacts and therefore the preregistered latency bound (median ≤ 60 s) could not be assessed.
- External validity: task corpus derives from public NetConfEval-style examples and a seeded synthetic generator limited to two vendor-dialect clusters; results do not imply universal benefit across other vendors, larger topologies, or production deployments.

VII. CONCLUSION

We executed a preregistered paired experiment that evaluated a GVR pipeline (gpt-5-mini + deterministic_netops_validator_v5 + ≤ 1 automated repair) on intent→CLI tasks under shared_initial_candidate semantics. Across 266 analysed pairs the guarded pipeline improved validator pass rate by 0.01879699248 but did not meet the preregistered exact-McNemar $\alpha=0.05$ decision rule ($p=0.0625$); a paired bootstrap CI ($B=10,000$, $seed=1729$) narrowly excludes zero. Repair calls were rare (6/300) and median repair iterations per task = 0. Per-episode latency was not archived and could not be assessed. All execution and analysis artifacts (raw responses, validator traces, provider call accounting, and reconciliation manifests) are archived in the evidence bundle to support reanalysis and follow-on work.

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DISCLOSURE STATEMENT

Disclosure (concise): Declared model: gpt-5-mini (execution/model_configuration.json archived; declared_model_version = "gpt-5-mini"). Orchestration adapter: hosted_netops_gvr_v1. Deterministic validator: deterministic_netops_validator_v5. Preregistration: cnsm2026-001-gvr-batfish-prereg-v1 (manifest SHA-256: 46a223492f760950b3d06f475129ce21e52e296c4c2a1e09df3ccd50bec9f891). Initial master prompt immutable reference: provenance/master_prompt.txt (SHA-256: 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582b39b15ba). Human role and autonomy boundary: before the autonomy lock a human Research Director selected the topic family and approved pre-lock infrastructure development; after the autonomy lock the registered pipeline autonomously performed literature selection, preregistration, experimental execution, analysis, manuscript drafting, AI peer review simulation, and revision. Nondeterminism disclosure: archived execution/model_configuration.json records temperature = null (unset) and no deterministic sampling seed is archived; therefore nondeterministic sampling of model outputs cannot be excluded for the recorded provider calls. Execution and analysis artifacts, logs, and hash manifests are archived in the evidence bundle referenced in the preregistration.

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